

HOW TO CHANGE THE INDUSTRY

The genetic engineering business is not a laboratory. It is a small number of firms, a body of law those firms helped write, and a handful of desks where the rules are actually decided. This is where those desks are, who sits at them, and how a private citizen in any country reaches them — starting from nothing and spending almost nothing.

You are told this is a question for scientists. Most of it is not. The decisions that set the terms are taken in a patent office, in a technical annex to a regulation, in a chapter of a trade agreement, and in a standing committee whose members are listed on a public website. Each of those places accepts documents from members of the public, and each of them runs on deadlines. Seed networks and farmers' organisations have used them to get patents struck out, to hold up a rule for years, and to write a planting ban into a national constitution.

This guide is about the industry and the law behind it, not about one release. If somebody has applied for permission to put an engineered organism in a field near you, that is a shorter and more urgent fight with its own clock. Deal with that first. What follows is the shape of the thing itself, then nine numbered places to reach it. The numbers match the badges on the map overleaf.

WHAT IS ACTUALLY BEING SOLD

Four different businesses travel under one name, and confusing them is the first way people lose an argument.

Seed and chemistry, sold together. Four firms — Bayer, Corteva, Syngenta and BASF — hold roughly 56 per cent of the world's commercial seed and 61 per cent of its pesticides. Bayer alone is about 23 per cent of the seed; Syngenta about 25 per cent of the pesticides. Competition authorities generally start calling a market distorted when the largest four hold 40 per cent of it. The dominant engineered trait is herbicide tolerance, which means the seed and the chemical it is built to survive are bought as a pair, from firms that sell both.

Animals, trees and insects. Gene-edited livestock, fast-growing fish, engineered forest trees and modi-

fied insects sit under weaker rules than crops in most countries, and often under no biosafety rule at all. Gene drives — constructs built to push themselves through a wild population — have not been released anywhere yet. Field trials are being planned for around 2030.

Industrial fermentation. Engineered yeast and bacteria in closed tanks making flavours, enzymes, fats, proteins and drugs. The fastest-growing part of the industry, largely outside biosafety law because nothing is deliberately released, and regulated instead through food additive approvals and novel food rules.

Medicine, and the edge past it. Gene therapy on a living patient is a treatment, and it is not what this guide is about. Editing an embryo passes the change to every generation afterwards. A few privately funded companies have said they intend to work on it, in jurisdictions chosen for their permissiveness. The 2023 international summit position was that preclinical safety is not established and the societal debate is not concluded.

THE RULES ARE NOT MADE WHERE YOU THINK

Almost nobody who follows this issue is watching the five places where the terms are set. None of them is a parliament. All five take submissions from ordinary people.

1. The definition. A sentence in a technical annex decides whether a whole class of organism is regulated at all. Argentina wrote gene-edited organisms out of its biosafety law in 2015; Chile, Brazil, Colombia, Paraguay, Honduras, Guatemala, El Salvador, Costa Rica and Uruguay followed. The United States, Canada and Australia let the developer make the determination itself. When the definition moves, the permit, the register, the comment period and the public record all disappear at once.

2. The patent office. European law forbids patents on plants obtained by conventional breeding, so applicants claim the gene variant instead. Patents on natural variation in spinach, tomato and lettuce were granted in 2025, with around forty further applications of the same kind filed that year. A patent examiner is not a minister and holds no vote, and the office still has to answer written objections from anyone who files them.

3. Seed law. The plant variety system decides what a farmer may lawfully do with a harvest. The 1991 text of the UPOV convention had 80 members covering 99 countries by mid-2025. Ghana's 2020 act carries a ten-year minimum sentence. Seventeen West African states are bound through a single regional office in Yaoundé. Saving, exchanging and selling seed is where this industry meets most of the world's farmers, and it is decided in a law that rarely mentions genetic engineering at all.

4. Trade agreements. Accession to UPOV 91 is written into trade deals as a condition, which converts a domestic argument into an international obligation before the domestic argument has happened. Mexico restricted genetically modified maize; a USMCA panel ruled against it in December 2024, and Mexico then wrote the planting ban into its constitution in March 2025.

5. The standing committee. The scientific panel that advises the regulator reviews a dossier written by the applicant and rarely runs experiments of its own. Its membership, its agendas, its minutes and its members' declared interests are usually published. Usually nobody reads them.

WHICH OF THE FOUR SYSTEMS YOU ARE IN

Four regulatory worlds exist, and the same letter sent into the wrong one gets no reply. Most countries are parties to the biosafety protocol and run a national biosafety act: an authority, a technical committee, a

public register, and a comment period written into the statute. European Union members run two tracks — national permits for trials, a union-wide procedure for market approval, a standing right for any member state to prohibit cultivation on its own soil, and from July 2028 a separate two-tier regime for the newer techniques. The United States sits outside the treaty and divides the work between three agencies under a framework built in 1986, taking comment through public dockets. And a growing group have written gene editing out of the definition altogether, which removes the permit and everything attached to it. Establish which one you are in before you write anything.

NAME THE LINE BEFORE YOU MOVE

"The industry has gone too far" is not a campaign. Different objections attach to different doors, and the first pedagogical act is to say precisely which one is yours.

■ **Ownership of seed.** Farmers prosecuted for saving a harvest, patents on varieties that already existed. Doors 3 and 4.

■ **Secrecy.** Assessments written by applicants, dossiers marked confidential, committees meeting without minutes. Doors 2 and 6.

■ **Irreversible release.** Gene drives, engineered trees, insects, soil microbes. Doors 1, 8 and 9.

■ **The chemistry attached to the trait.** More spraying, and a separate approval process for the spray. Doors 5 and 7.

■ **Public money, private ownership.** Publicly funded research patented and licensed exclusively. Door 5.

■ **Consent.** Releases and trials on the land of people who were never asked. Doors 7 and 9.

■ **The germline.** Editing that is inherited. Doors 6 and 9, and a medical regulator rather than an agricultural one.

THE FIELD — EVERY PART OF IT, AND WHERE IT SITS

NINE DOORS · THE BADGES MATCH THE NUMBERED STEPS IN THE TEXT

THE LAW ABOVE EVERY COUNTRY

what your government already signed

WHAT FEEDS IT

four inputs — and four public doors in

THE FIRMS

and the bodies that speak for them

WHERE THE RULES GET WRITTEN

none of these is a parliament

WHAT COMES OUT

four businesses, one name

WHO CARRIES THE RESULT

the ground under all of the above

WHERE YOU PUSH

all pressure runs upward

CARTAGENA PROTOCOL

173 parties. Consent before import. Uncertainty is a lawful reason to refuse. Social and economic effects count.

THE DEFINITION

One line in a technical annex. If the organism is not “regulated”, the permit, the register and the comment period vanish together.

THE BIG FOUR

Bayer · Corteva · Syngenta · BASF. About 56% of world seed and 61% of pesticides between them.

ANNEXES & IMPLEMENTING ACTS

Where a skeleton regulation gets its flesh. Published for comment, on a clock. Hardly anyone comments.

SEED + CHEMISTRY PACKAGE

Herbicide tolerance is the dominant trait. The seed and the spray are sold as a pair.

FARMERS & SEED SAVERS

Saved seed, licence contracts, debt, and in some countries a criminal penalty.

YOU + THE LOCAL COALITION

Farmers, beekeepers, certifiers, exporters, a scientist, the union, the co-op, the parish.

AARHUS · ESCAZÚ

You must be consulted on these decisions. A compliance committee takes complaints from any citizen, free.

PATENT OFFICES

Monopolies over gene variants that already existed. Any person may file written observations before grant.

STARTUPS & INSTITUTES

Gene drives, engineered animals and trees, fermentation, embryo editing at the fringe.

THE STANDING SCIENCE PANEL

Reviews the applicant's file, rarely runs tests of its own. Membership and interests are published.

ANIMALS, TREES, INSECTS

Weaker rules than crops, or none. Gene drives are not yet released anywhere.

WILD RELATIVES & CENTRES OF ORIGIN

Where the ancestors of a crop still grow, one crossing is permanent.

LOCAL & REGIONAL GOVERNMENT

Zones, land use, water, procurement, extension, public land, community seed banks.

NAGOYA-KL LIABILITY

In force 2018. Who pays and who repairs when a release causes damage. Almost no national file mentions it.

SEED & TRADE LAW

UPOV 91 across 99 countries, carried into trade deals. Decides what a farmer may do with a harvest.

TRADE ASSOCIATIONS

Where the lobbying money hides. Registers, minutes and declared interests are public.

THE SIGNING DESK

National biosafety authority with the agriculture, environment and health ministries.

FERMENTATION IN TANKS

Enzymes, fats, proteins, flavours. Nothing is released, so biosafety law rarely applies.

INDIGENOUS & LOCAL PEOPLES

Consent here is a legal right, not a courtesy that was skipped.

COURTS

They rule on procedure, not on wisdom — and only ever as well as the record you made.

UNDRIIP · ILO 169 · PLANT TREATY

Consent before measures on indigenous land. Farmers' right to save, exchange and sell seed.

PUBLIC MONEY

Research grants, universities, subsidy, extension, development banks, public pension funds.

THE APPLICANT'S DOSSIER

The risk assessment is written by the party asking for the permission.

TRADE PANELS & STANDARDS BODIES

WTO notifications and dispute panels, harmonisation working groups, food standards codes.

MEDICINE, AND THE GERMLINE

Treatment of one patient, or a change that every generation afterwards inherits.

NEIGHBOURS, BUYERS, EATERS

Drift and contamination, lost certification, export contracts, labelling.

OUTSIDE THE COUNTRY

Treaty meetings, the Aarhus committee, UN rapporteurs, importers and certifiers.

ALL PRESSURE RUNS UPWARD

How to read this. Start at the bottom. You and whoever will stand with you are at the foot of it, and every route upward is drawn. The instruments your government already signed sit across the top and press down on all of it; the distance between what was signed and what was implemented is the gap you work in. The second band is what feeds the industry, and each of those four inputs has a public door into it. The firms sit in the middle, assessed largely on their own paperwork. Below them are the four desks where rules are actually written — none is a parliament, and all four take documents from members of the public. Trade law pushes back from outside the frame: a national ban can be attacked as a barrier to trade, which is why constitutions, court judgments and a thousand municipal declarations hold where a decree does not. The nine badges are the nine doors in the text. Find the one nearest the thing you object to, and start there rather than at door one.

1 ESTABLISH WHETHER THE LAW APPLIES AT ALL

Everything else depends on the answer, because it decides whether you are working inside a procedure or against a rule. Send the national biosafety authority a single question: does our biosafety law cover this class of organism, and if it does not, which instrument took it out? File the reply. It costs a stamp and it underpins every letter you write afterwards.

Three answers are possible. It is regulated, and there is a permit file, a register and a comment period. It was regulated and has been carved out, in which case find the instrument that did the carving — a resolution, a guidance note, a ministerial decree — and read how it can be amended, because that is now your target. Or nobody answers, which is itself a finding, and you appeal it under your access-to-information law.

The European Union adopted a regulation on new genomic techniques in July 2026, splitting them into two tiers, applying from July 2028, with the implementing detail still being written. Across Africa a growing number of countries — Nigeria, Kenya, Ghana, Malawi, Ethiopia, Burkina Faso among them — have issued genome-editing guidance rather than legislation. Guidance is easier to change than a statute, in both directions.

2 GET INSIDE THE RULE WHILE IT IS STILL BEING WRITTEN

A regulation is a skeleton. The flesh is added afterwards in implementing acts, technical guidance, annexes and detection protocols, and that is where the questions that matter get settled: what counts as equivalent to conventional breeding, whether products must be labelled, whether a method of detection has to exist before approval, who monitors and for how long. These drafts are published for comment, on a clock, and hardly anyone comments.

Where to find them. Your ministry's consultation page and the official gazette. The EU's Have Your Say portal, which is open to residents of any country, for the implementing acts running from 2026 to 2028. In the United States, dockets at regulations.gov: the agriculture department's rules on engineered plants were vacated by a federal court in December 2024, sending

the framework back to its earlier form, and rewriting has been under way since.

The one hardly anybody uses. Before a member government can enforce a new technical regulation, it has to notify it to the World Trade Organization, and those notifications are published in a free searchable database with the comment period attached. It is often the earliest public sight of a draft rule, and it covers countries whose own consultation pages are decorative. Search it by country and by product, and set an alert.

A SUBMISSION THAT COUNTS — FIVE PARTS

First line. The consultation reference, the instrument, and the paragraph number you are answering.

What you are. Grower, beekeeper, certifier, importer, doctor, resident of the district. State what you stand to lose by name. Standing in a later case is built out of this sentence.

One defect that can be checked. "The draft requires no validated detection method, which makes the labelling obligation in Article 12 unenforceable."

What it breaks. The article of national law, of the Cartagena Protocol, or the constitutional right your own courts already recognise.

What you want done. Amend this paragraph in this way. Ask for something narrow enough that granting it costs them little.

3 THE PATENT OFFICE, WHICH ANYONE MAY WRITE TO

Patents are granted by examiners working from the applicant's own text, under time pressure, and the law of every major patent system builds in a way for outsiders to interfere. Two of them are worth knowing.

While an application is still being examined, any person may file written observations on why it should not be granted. There is no fee, no requirement to be affected, and no obligation to become a party to the proceedings. At the European Patent Office this is Article 115; most national offices have an equivalent. India goes further and allows any person to oppose before grant, and excludes plants, animals and seeds from patentability outright.

After grant, European patents can be opposed for nine months by anyone at all, on grounds including lack of novelty and unpatentable subject matter. Opposition carries a fee and takes years, and it is generally run by coalitions rather than individuals. The No Patents on Seeds coalition has used it repeatedly; an opposition to a maize patent was rejected in November 2025 and is under appeal.

Public opinion is not the obstacle here. Surveys in Germany, France, Italy, the Netherlands and Poland have put opposition to patents on seed at around 80 per cent. A European Parliament vote to ban such patents in February 2024 was dropped from the negotiated text in December 2025. The provision was removed in a closed negotiation, and that is the kind of fact worth putting in front of the members who voted for it.

What to look for in a patent. Read the claims, not the title. Ask whether the claim covers a plant that a conventional breeder could produce, whether it covers a variation that already exists in nature, and whether it reaches the harvest and the next generation of seed. Those three questions are the ones examiners answer badly.

4 SEED LAW, WHICH IS WHERE MOST FARMERS MEET THIS

Plant variety protection law is drafted by specialists, passed with little debate, and then decides whether a farmer commits an offence by keeping seed. Watch for a bill implementing UPOV 91, a criminal penalty attached to seed exchange, a compulsory variety register that makes unregistered local seed unlawful to sell, and a certification scheme written so that only uniform commercial varieties can pass it.

You have a counterweight that is rarely cited. The International Treaty on Plant Genetic Resources for Food and Agriculture recognises farmers' rights to save, use, exchange and sell farm-saved seed, and leaves it to national governments to protect them. Most governments have done nothing to implement it. That gap is the argument: your country signed one instrument protecting farmers' seed and is now drafting another that criminalises it.

The regional level matters more than it looks. Seventeen West African states legislate through a shared intellectual property office, so a change there binds all of

them at once, and a campaign there reaches all of them at once. Pan-African seed governance conferences have been running since 2023 and produce declarations that national delegations then have to answer for. Trade negotiations with the European Union have carried UPOV 91 clauses into talks with the Philippines and with southern and eastern African states.

5 FOLLOW THE PUBLIC MONEY

A great deal of this industry runs on public funds and public institutions, which is a weakness, because public money comes with public documents.

■ **Research grants.** Ask which public bodies fund engineering work in your country, on what terms, and who owns the resulting patent. Grant agreements are usually obtainable. Publicly funded, exclusively licensed is the pattern to look for.

■ **Universities.** Technology transfer offices publish licensing policies and often annual reports naming the licensees. A university senate is a body that can be lobbied by twenty people, which is not true of a ministry.

■ **Subsidy and extension.** Agriculture ministries distribute seed, credit and advice. If the public extension service only demonstrates one kind of seed, that is a procurement decision and procurement decisions have rules, appeals and published criteria.

■ **Development banks and aid.** Multilateral development banks fund agricultural programmes, and each operates an independent complaints mechanism that accepts submissions from affected people, without a lawyer, in their own language. Use the bank's own environmental and social standards as the yardstick.

■ **Public pension funds.** They hold shares in these firms, they answer to a board, and their board meetings are usually open. A resolution filed by shareholders forces a written answer from the company.

6 READ THE LOBBY REGISTER AND THE DECLARATIONS OF INTEREST

This is the least glamorous door and it produces the most usable material. Bayer's own declaration for 2022 put its European lobbying at around six and a half million euros, its worldwide spending at around forty-nine million, and its association membership fees at a further twenty-six million. Trade association spending is where the money hides, because the association lobbies and the member firm's own declaration stays modest.

Collect four things and keep them in one folder. The lobby register entry for each firm and each association active in your country. The membership list of the official advisory committee, with each member's declared interests. The minutes of meetings between the ministry and the industry, which in many systems must be released on request. And the career histories: officials who left the regulator for the industry, and industry staff who joined the regulator.

None of this proves corruption and you should not claim it does. What it does is remove the assumption of neutrality from the room, which changes how a journalist, a parliamentary committee and a court read everything else you file.

7 LOCAL GOVERNMENT, WHERE TWENTY PEOPLE STILL COUNT

A ministry is lobbied professionally every week of the year, and a hundred letters do not register there. A district or municipal council is a room with a dozen people in it, and that dozen controls things no ministry touches: land use, water, roads, school and hospital purchasing, the agricultural extension service, public land, and whatever seed the council itself distributes. Regions and municipalities across Europe, Latin America and Africa have used those powers to declare themselves free of engineered cultivation, and the declarations have outlasted national approvals.

Assemble the group before you need it, out of people with something concrete at stake: growers who keep their own seed, beekeepers, a certifier, an exporter whose contract specifies non-engineered stock, the irrigation association, a technician from the agricultural college, and whoever in the district already gets people

into a room — a co-operative, a union branch, a church, a market committee. Lead with the export argument when you deal with agriculture officials. Lost contracts arrive in a currency they already keep accounts in.

The constructive half matters as much as the refusal. A council that funds a community seed bank, buys local varieties for its own institutions and pays an extension worker who knows them is building the thing that makes the refusal survivable. Opposition that offers farmers nothing is opposition that loses farmers.

8 COURTS, WHICH RULE ON PROCEDURE AND NOT ON WISDOM

No court is going to decide whether genetic engineering is a good idea. Courts decide whether the decision-maker kept to the procedure the statute lays down, and at the industry level that means going after a rule rather than a permit. The grounds that work are the same everywhere: the agency exceeded its statutory powers, it failed to assess environmental effects when the statute required it, it consulted nobody, it gave no reasons, it relied on an assessment written by the party that stood to gain.

In December 2024 a United States federal court vacated the rule governing engineered plants, on the ground that the agency had not done the environmental analysis the law required. In the Philippines, an appeal court revoked the biosafety permits for two crops in April 2024, in a 143-page judgment resting on the precautionary principle and on the absence of any monitoring mechanism. The government appealed and the litigation has continued ever since. Plan on that timescale. In India, a split verdict in July 2024 established that the national approval committee's decisions are judicially reviewable and ordered the government to produce a national policy.

Go to a university environmental law clinic or a public-interest firm before you go to a commercial one. The Environmental Law Alliance Worldwide will introduce you to a lawyer inside your own country, in more than seventy of them, and charges nothing for the introduction. In the Aarhus countries there is an article requiring that environmental cases not be prohibitively expensive. Cite it in writing at the start, not when the bill arrives.

9 OVER THEIR HEADS

When the national door shuts, the file becomes evidence somewhere else.

■ **The treaty meetings.** The governments party to the biosafety protocol meet every two years and settle the rules their own authorities then apply. The next round is in Yerevan, 19–30 October 2026, with synthetic biology and digital sequence information on the agenda. Your country is sending a delegation, and delegations travel with written instructions drafted at home. Reach them while they are still at home, with one specific request.

■ **The Aarhus Compliance Committee.** Any person may lodge a communication against a party government. No fee, no lawyer, no nationality test. It moves slowly, it works in public, and governments would rather it did not exist.

■ **UN special rapporteurs.** The mandates on toxics, the right to food, indigenous peoples' rights and environmental defenders take submissions through an online form, from anybody.

■ **Buyers and certifiers.** A purchaser who refuses a consignment, or a certifier who withdraws a mark, reaches the firm at a place no ministry can cushion for it.

The instruments your government already signed sit behind all of this: the biosafety protocol with 173 parties, which permits a government to weigh social and economic effects and not only laboratory risk; the liability protocol in force since 2018, which almost no national file mentions; and the consent requirements in the indigenous rights declaration and in labour convention 169. The gap between what was signed and what was implemented is the most reliable argument in this entire field.

ARGUMENTS THAT SURVIVE CONTACT

Be ruthless here, because a bad argument does not merely fail. It discredits everyone standing next to you and gets used against your side for years.

The ones that lose. That eating it will harm you: the authority keeps a thick folder of feeding studies for exactly that question and will put it on the table. That it is unnatural: no authority anywhere is obliged to act

on the word, and using it hands the definition to the other side. Leave both alone. Neither is where this industry is weak.

The ones that hold.

■ **Concentration.** Four firms, most of the world's seed and most of its pesticides, selling the trait and the chemical as a pair. This is a competition argument, and competition authorities have powers ministries do not.

■ **Patents on things that already existed.** Natural gene variants claimed as inventions, and the claim reaching the harvest and the seed after it.

■ **Criminalised seed.** A farmer facing a prison sentence for keeping part of a harvest.

■ **Who wrote the assessment.** The applicant did, and the panel reviewed the applicant's file.

■ **No liability.** Put it in writing — if this causes loss, who compensates whom, under which provision? Then file the non-answer, which is the useful document.

■ **No monitoring.** Nothing in the file says who measures what happens afterwards, on what schedule, or for how many years.

■ **Irreversibility.** An organism that reproduces cannot be withdrawn from the market the way a batch of a chemical can. No regulator has a procedure for it, and the files show that they do not.

■ **Public money, private patent.** Funded by the state, owned by a firm, licensed exclusively.

■ **Consent.** Which body agreed, on whose behalf, on what date, and on the basis of what information.

WHAT THEY DO BACK

■ **They redefine.** Reclassification is cheaper for them than refutation. Move the organism outside the definition and there is no permit to contest, no consultation to join and no register to read. Give the annexes the same attention you give the applications.

■ **They use trade law.** A national restriction can be challenged as a barrier to trade, and the challenge sometimes wins. Decrees fall most easily. Constitutional clauses, court judgments and several hundred separate municipal decisions are far more expensive

- to dismantle.
- **They rebrand.** Every technique arrives as a solution to hunger, drought or climate. Ask what is actually planted at scale today, and compare the answer to the brochure.
 - **They wait you out.** A file of this kind moves over five to fifteen years and volunteer groups rarely last three. Settle at the beginning who holds the archive, and keep a duplicate with somebody outside the group.
 - **They separate you from the farmers.** Opposition that arrives from outside and tells growers what they may plant loses, and ought to. The people with land at stake belong in the room on the same footing as everybody else.

NINE WAYS PEOPLE LOSE THIS FIGHT

1. Attacking a product and leaving the rule that authorised it untouched. 2. Opening with food safety, and spending the credibility you needed for everything after it. 3. Letting a consultation close, then finding out that the courtroom depended on it. 4. Accepting a confidentiality claim without appealing it. 5. Keeping the whole archive on one person's laptop, and that person leaving. 6. Speaking on behalf of growers or an indigenous authority instead of making sure they were asked directly. 7. Signing a non-disclosure agreement in order to attend a briefing, and losing the right to describe it afterwards. 8. Briefing a newspaper before the documents are in hand, so it becomes one opinion against another. 9. Saying corruption when what you have is a suspicion, with no register entry, no minutes and no dates.

WHAT HAS ACTUALLY MOVED

Neither despair nor optimism is much use. Look at the record, both columns of it.

Against the industry. India held public hearings on an engineered aubergine and imposed a moratorium in February 2010 which has never been lifted. A Philippine appeal court revoked the biosafety permits for two crops in April 2024. Mexico lost a trade case in December 2024 and then wrote the planting ban into its constitution three months later. A United States federal court struck down the rule governing engineered plants in December 2024. In Burkina Faso a na-

tional coalition spent years documenting an engineered mosquito programme; after a release in August 2025 the judicial police sealed the insectary and the government suspended the project nationwide inside a fortnight, on the strength of a file that had been building for years.

For the industry, over the same years. Nine countries in Latin America wrote gene editing out of their biosafety law. The plant variety convention reached 99 countries. Patents on natural gene variation were granted in Europe in 2025. A parliamentary vote to ban seed patents was removed from the text in a closed negotiation.

Both columns are real. What the wins have in common is that they were procedural, or local, or judicial, and that none of them was produced by one spectacular act.

YOUR FIRST MONTH

Week one — find out what you are in. Write the one-sentence question at door 1. Find your national biosafety register and the biosafety clearing-house entry for your country. Write down the name of the authority, the name of the advisory committee, and the date of its next meeting.

Week two — collect the paper. The lobby register entries. The committee membership and declared interests. Any draft rule out for consultation, with its closing date. One patent application covering a crop grown where you live. Keep all of it in one shared folder with two other people holding access, on the assumption that any one of you may stop.

Week three — find the five people who lose something. The grower who keeps seed, the beekeeper, the certified exporter, the community whose claim covers the district, the technician at the agricultural college. Ask each of them what they have already tried. Most of these arguments have been run somewhere before, and the people who ran them will say which door opened.

Week four — file one thing, on one defect. One submission, one checkable defect, one named provision, one specific remedy, sent by a method that produces a receipt. Have somebody else file separately on a different defect. Ten people filing ten defects will

outwork a petition carrying a thousand names.

SEVEN WORDS THAT DO THE WORK

Deliberate release. Putting an engineered organism outdoors where it can live, move and reproduce. The legal term in most biosafety laws.

Unique identifier. The standard code assigned to one engineered line, which follows it through every database in the world. The most useful string you will ever type into a search box.

Deregulation. A decision that the organism may be grown without asking permission again.

New genomic techniques. The newer editing methods. Whether they count as engineering in law is the single most contested question in this field.

UPOV 91. The 1991 plant variety convention. It restricts what a farmer may do with protected seed.

Coexistence. The official word for rules meant to keep engineered and unengineered crops apart. Read the file to see whether any exist.

Free, prior and informed consent. The standard that applies before measures affecting indigenous peoples' land, under UNDRIP and ILO 169.

WHERE TO LOOK, AND WHO ANSWERS STRANGERS

Documents. The Biosafety Clearing-House at bch.cbd.int for national decisions and focal points; your national biosafety register; the EU GMO register and the food safety authority's application register; the agriculture department's petitions pages and the dockets at regulations.gov in the United States; Espacenet and Google Patents for patent families; the WTO technical notification database for draft rules; your national lobby register.

Organisations that reply to strangers. The Environmental Law Alliance Worldwide, for a lawyer in your own country. Third World Network's biosafety programme, for what is moving across the global South. Testbiotech and GMWatch, for dossier and patent analysis. The African Centre for Biodiversity. GRAIN and the No Patents on Seeds coalition on seed ownership. And your national organic certifier association, which carries the financial risk of contamination and answers faster than a ministry.

LETTERS THAT GET ANSWERED

One reference number in the first line, and one question in the body. A letter asking six things gets a reply addressing none of them, so ask the one whose answer has to be a document or a date. Name the provision that obliges them to reply and the number of days it allows them. Copy someone in — the environment ministry when you have written to agriculture, the ombudsman, the information commissioner, the biosafety focal point. Use a method that generates a receipt, and keep it. Number your letters. Three unanswered letters, numbered and dated, is itself a publishable fact.

IF YOU DO ONLY ONE THING THIS MONTH

Find the consultation that is open in your country now, and write down the day it closes where somebody other than you can see it. Every other task here survives being late. This one does not: most legal systems will not hear an objection in court that was never put during the consultation, so the closing date shuts the courtroom at the same time as the file.